# Immunity and vaccination

*9C Threat from disease · Lesson 2 of 7*

Date _____ Class _____

**Today I can**

- ☐ I can describe how phagocytes and lymphocytes destroy microorganisms in the body.
- ☐ I can explain why you only get many infectious diseases once, and how vaccination protects people and communities.

**What happens today**

1. Do Now: once is enough
2. Learn: Two kinds of white blood cell · Why you are ill before you get better · and more
3. Core tasks
4. Challenge, then the exit ticket

Student Book pages 39 to 41 · Workbook pages 29 to 30

**What I learned today (one or two sentences)**

# Key Vocabulary

| Term | Definition |
|---|---|
| **white blood cell** | a blood cell that fights pathogens |
| **phagocyte** | a white blood cell that surrounds, ingests and digests microorganisms |
| **lymphocyte** | a white blood cell that makes antibodies |
| **antibody** | a protein that sticks to one type of pathogen |
| **specific** | fits only one type of pathogen |
| **memory cell** | a lymphocyte that stays in the body ready to make antibodies quickly |
| **immune** | protected against a disease, so you do not become ill with it |
| **vaccine** | a substance containing parts of a pathogen, given to make a person immune |
| **virus** | a very small pathogen that can only copy itself inside a cell |
| **dose** | the amount of a medicine given at one time |
| **half-life** | the time for the amount of a medicine in the body to fall by half |

# Quick Review

*once is enough*

Many people had chickenpox as a small child. If a friend has it now, they will almost certainly not catch it again.

**1** Suggest why you only get chickenpox once.

**2** What is a pathogen? (last lesson)

**3** Name two things your body does to fight an infection.

# Core 1 of 3

*Everyone does these. Answer on the card or under it.*

**1** Complete the text using words from the word bank. Some words are used more than once.

When you are infected by a _____, white blood cells start to attack.

_____ ingest some of the microorganisms, while _____ start to produce

_____ that stick to the outside of the microorganisms. These proteins may cause the

_____ to burst open, or cause several microorganisms to _____ together,

which makes it easier for them to be _____.

# Core 2 of 3

*Everyone does these. Answer on the card or under it.*

**2** Match each key word to its meaning. Draw a line.

| | |
|---|---|
| phagocyte | a lymphocyte that stays in the body, ready to make antibodies quickly |
| lymphocyte | a protein that sticks to one type of pathogen |
| antibody | a white blood cell that ingests and digests microorganisms |
| memory cell | contains parts of a pathogen and makes a person immune |
| vaccine | a white blood cell that makes antibodies |

# Core 3 of 3

*Everyone does these. Answer on the card or under it.*

**3** Tick one box in each row to show whether the statement is true or false.

| Statement | True | False |
|---|---|---|
| You are ill before you get better because new antibodies take days to make. | | |
| One type of antibody fits every type of pathogen. | | |
| A vaccine gives you a mild case of the disease. | | |
| Memory cells make the second response faster and larger. | | |
| Vaccinating most people can stop a disease spreading in a community. | | |

# Challenge 1 of 3

*When Core is finished. Full sentences.*

**C1** A vet uses an antibiotic called ceftiofur. The table gives the dose and the half-life for some animals. The half-life is the time for the amount of antibiotic in the animal's body to fall by half.

| Species | Dose (mg of antibiotic per kg of animal) | Half-life (hours) |
|---|---|---|
| alpaca | 6.6 | 45 |
| American black duck | 6.6 | 32 |
| ball python | 10 | 64 |
| goat | 15 | 37 |

**a)** State what is meant by the dose of a medicine.

# Challenge 2 of 3

*When Core is finished. Full sentences.*

**C1** *continued*

**b)** To be effective, the concentration of ceftiofur in a ball python must stay above 2.5 mg/kg. Calculate how many days after the first dose a second dose is needed. Show your working.

**c)** Calculate the dose, in mg, for a 15 kg goat.

# Challenge 3 of 3

*When Core is finished. Full sentences.*

**C2** Explain how vaccinating most of the people in a community protects people who cannot be vaccinated, such as newborn babies.

# Exit Ticket

*Hand this card in at the end of the lesson.*

Name _____

**1** Name the two types of white blood cell and say what each one does.

**2** Why do you only get measles once?

**3** What does a vaccine contain, and why does it not give you the disease?

**Today I can:** Goal 1 ☐ Goal 2 ☐